University of BRISTOL

Angela Attwood <psasa@bristol.ac.uk>

RE: Request for information: OSF's replication of Farris et al. (2008)

1 message

Farris, Coreen <cfarris@rand.org> 8 January 2015 at 17:38
To: Angela Attwood <Angela.Attwood@bristol.ac.uk>

Right (except for the inverse part). So the full calculation would be:

Ln(α) for Fiendliness = ln ((63/70 / 4/70) x (63/70 / 2/70) x (63/70 / 1/70))

———————————————————————

(√2*4) * (4-1)

= 0.609877

A couple of notes:

1. We used the Ln transformation instead of the log transformation (from Macmillian and Creelman). I can't remember the reason, but it's documented in the paper. Probably to make calculations more tractable.

2. For participants who have zeros for some response categories (e.g., someone who doesn't identify any of the friendly pictures as sad = 0/70), you'll run into division by zero problems in the equation and it won't produce a sensitivity value. There are a number of recommended corrections in the literature. We chose to use the simplest, which is to replace zeros with a value that is one-half the size of the next smallest response (1). In short, replace zeros with 0.5 so that instead of 0/70=0%, you'll use 0.5/70=0.7% I can dig out the citation for the imputation if you need it, but it may take some time. It's been awhile since I worked on this.

From: Angela Attwood [mailto:Angela.Attwood@bristol.ac.uk]
Sent: Thursday, January 08, 2015 12:06 PM
To: Farris, Coreen
Subject: Re: Request for information: OSF's replication of Farris et al. (2008)

Sooo….

For any one sensitivity extraction we just need the following data (in this case stimulus is friendly):

Response Friendly: 63 (i.e hit)

Reponse SI: 4

Reponse Sad: 2

Respons Rej: 1

My calculation for numerator is the inverse of: (63/70 / 4/70) x (63/70 / 2/70) x (63/70 / 1/70)

where correct hits are represented on in first part of each ratio (63/70) and misses broken down by category make up second part (i.e., participant responds sad to friendly stimulus for example). Is this right?

MANY thanks!

On 5 January 2015 at 15:11, Farris, Coreen <cfarris@rand.org> wrote:

Hi Angela –

I've attached a more thorough explanation for how to calculate sensitivity with multiple choice designs (from Macmillan & Creelman, 1991, the *Choice Theory Analysis* section). You may find it helpful.

Your attached explanation looks like you're on the right track, but missing two pieces. First, the product sign signifies that you'll need to compute the ratio in the numerator across all categories and then multiply them. That will give you the sensitivity estimate for that particular category. Do the same for each of the three remaining categories, and then the four resulting sensitivity estimates can be included in a GLM to assess gender on positive-affect (friendliness and SI) and negative-affect (sad and rejecting) sensitivity.

Let me know if you have additional questions. Or feel free to call 412-683-2300, x4482. I'm on EST.

Coreen

From: Angela Attwood [mailto:Angela.Attwood@bristol.ac.uk]
Sent: Monday, January 05, 2015 7:47 AM
To: Treat, Teresa A; viken@indiana.edu; Farris, Coreen
Cc: reproducibilityproject@gmail.com
Subject: Re: Request for information: OSF's replication of Farris et al. (2008)

Dear Authors,

Re: Replication of Farris et al. (2008).

Many thanks again for your help with the replication study. I can confirm that we have finished testing and have also destroyed the task images.

I am contacting you now with a query regarding the calculation for the sensitivity scores. Our replication remit was that we only follow up one aspect of your findings so this is the only score we need to generate. However, I want to verify that I understand the formula correctly and that I do not miscalculate the scores. In particular, could you please confirm exactly what the formulaic notation is referring to - please see attached word document for more details.

Many thanks and very best wishes. I look forward to hearing from you but please feel free to contact me if you require any further information.

Angela

On 9 September 2014 at 02:18, Treat, Teresa A <teresa-treat@uiowa.edu> wrote:

Dear Dr. Attwood,

Thank you for checking with us about the study. Rick Viken and I are embedding responses to your questions below.

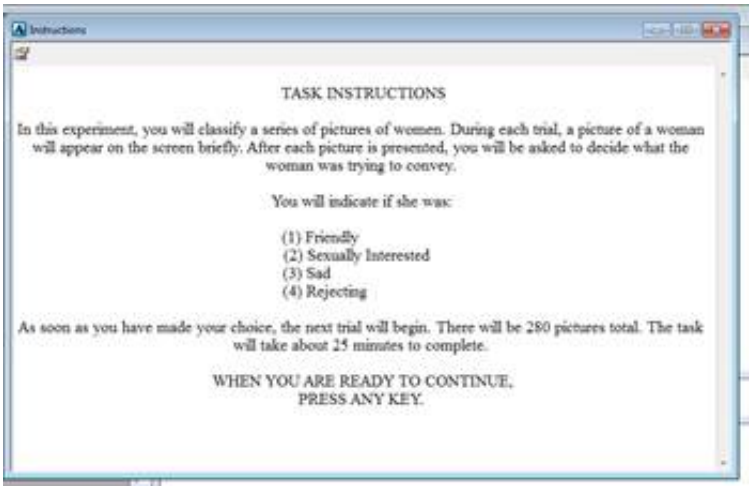
We have finished preparing the study and have drafted a methods section describing the study design and sampling plan. If you have a moment to review it, I would appreciate any comments that you have about the design -- particularly if you see anything that would interfere with replication. We want to do everything that we can to maximize the accuracy of our replication attempts. Do you see anything about the design that makes you anticipate that the original finding would fail to replicate? If so, what is it? And, are these suspicions based on published evidence about the constraints on the effect, things that you have identified in research but not published, or are they based on your informed judgment without yet having empirical evidence?

We do not have any concerns about factors potentially jeopardizing replication beyond those we already have articulated regarding shifts in participants' interpretations of the stimulus dimensions over time.

The paper was very clearly and comprehensively written and we do not have many additional queries. However, we have a few questions regarding fine-grained task elements which have arose during our programming of it, and which I wondered if you would kindly answer:

1. - Were any standardised instructions given to participants? Were these programmed into task?

Here are the instructions, as they were programmed into the task:

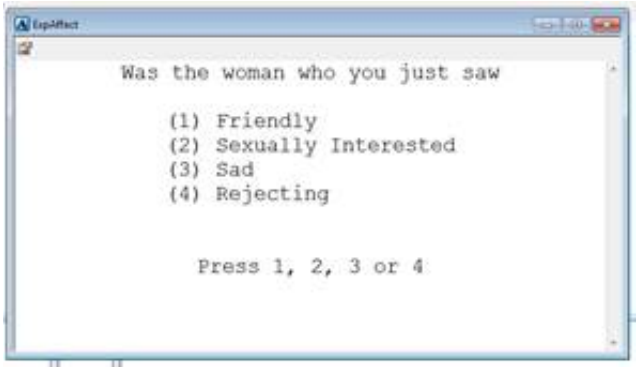




2. - Did each trial begin with a fixation cross and if so how long did this stay on screen?

Yes. 250 ms.

3. - The four descriptors appear together on screen – were they arranged linearly/square etc, and did the order change/stay same?



This screen appeared after each photo.

4. - How did participants respond? Mouse click on descriptor?

Participants pressed 1, 2, 3, or 4.

5. - Was there an interval between trials or did next trial start directly on response of previous?

The next trial started immediately.

We also read the preliminary paper you attached, in which you indicated that the materials would be available on the Open Science Framework. As discussed in our previous correspondence, the photos are available only for this particular replication, are not to be used for any other study or purpose, and are to be destroyed thereafter. Thus, we wanted to confirm that this understanding still is in place.

Best wishes --

Teresa Treat and Rick Viken

--

Dr Angela Attwood
Tobacco and Alcohol Research Group

MRC Integrative Epidemiology Unit
UK Centre for Tobacco and Alcohol Studies
School of Experimental Psychology
University of Bristol
12a Priory Road
BRISTOL BS8 1TU
United Kingdom

T: +44.117.3317814
F: +44.117.9288588
E: angela.attwood@bristol.ac.uk

http://www.bris.ac.uk/expsych/people/angela-s-attwood/
http://www.bris.ac.uk/expsych/research/brain/targ/

Follow us on Twitter: @TARGBristol and @AngelaAttwood

--

Dr Angela Attwood
Tobacco and Alcohol Research Group

MRC Integrative Epidemiology Unit
UK Centre for Tobacco and Alcohol Studies
School of Experimental Psychology
University of Bristol
12a Priory Road
BRISTOL BS8 1TU
United Kingdom

T: +44.117.3317814
F: +44.117.9288588
E: angela.attwood@bristol.ac.uk

http://www.bris.ac.uk/expsych/people/angela-s-attwood/

http://www.bris.ac.uk/expsych/research/brain/targ/

Follow us on Twitter: @TARGBristol and @AngelaAttwood

This email message is for the sole use of the intended recipient(s) and may contain confidential information. Any unauthorized review, use, disclosure or distribution is prohibited. If you are not the intended recipient, please contact the sender by reply email and destroy all copies of the original message.

--

Dr Angela Attwood
Tobacco and Alcohol Research Group

MRC Integrative Epidemiology Unit
UK Centre for Tobacco and Alcohol Studies
School of Experimental Psychology
University of Bristol
12a Priory Road
BRISTOL BS8 1TU
United Kingdom

T: +44.117.3317814
F: +44.117.9288588
E: angela.attwood@bristol.ac.uk

http://www.bris.ac.uk/expsych/people/angela-s-attwood/
http://www.bris.ac.uk/expsych/research/brain/targ/

Follow us on Twitter: @TARGBristol and @AngelaAttwood